

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Department of Electrical Engineering
2. Subject Name: NETWORK ANALYSIS (EE21203) L-T-P: 3-1-0 Credits: 4
3. Pre-requisites:
4. Syllabus and reference books:

Syllabus:

This is an introductory subject of Linear Time-Invariant Network Analysis.

Reference Books:

- 1) Network Analysis by M. E. VAN VALKENBURG, Pearson Education India

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Introduction – Development of the circuit concept, charge and energy, relationship between field and circuit, circuit parameters, conventions.	1
2.	Network Equations – Kirchoff's laws, source transformation, loop and node analysis, duality.	2
3.	Network Theorems – Thevenin's and Norton's theorem, Superposition and Reciprocity theorem, Maximum power transfer theorem, Millman's theorem, Tellegen's theorem, Inclusion of dependent source.	4
4.	Network Transients – Solution of networks represented by first order and second order differential equations, Inclusion of initial condition.	6
5.	Coupled circuits - Magnetic coupling, dot convention	2
6.	Circuit representation Analogous Systems – Mechanical systems (e.g., gear train, coupling, lever), D'Alembert's principle, Electro-mechanical system, Water level.	2
7.	Graph Theory – Graph of network, Trees, Loops, Incidence matrix, Cut-Set matrix, Derivation of relationships, Analysis of networks.	6
8.	Network under Non-Sinusoidal Input - Steady-state response of a network to non-sinusoidal periodic inputs, Power factor, Effective values.	2
9.	Laplace Transform and Its Use in Network Solution - Region of convergence, Properties of the Laplace Transforms, The Unilateral Laplace Transform and its examples and properties, Inverse Laplace Transform,	8

	Initial- and Final-value Theorems, Partial Fraction expansion, Heaviside's theorem, Singularity functions, Waveform synthesis, Analysis of RC, RL, and RLC networks with and without initial conditions using Laplace transforms, Evaluation of initial conditions.	
10.	Network Functions & Two Port Network Parameters – Network functions for One-Port and Two-Port networks, Poles & Zeros of Network Functions, Time domain behavior from Poles & Zeros, Open Circuit Impedance parameters, Short Circuit Admittance Parameters, Transmission Parameters, The Hybrid Parameters, Combination of Two-Port Networks.	5
11.	Analog Filters – Introduction to Low Pass, High Pass, Band Pass and Band Reject filters, Brief mention of ideal Operational Amplifier, Second-Order Low Pass and Band Pass filters (e.g., Sallen-Key), Butterworth Filter – Butterworth response – Pole locations – Low Pass filter specification, Chebyshev Filter – The Chebyshev Polynomial – Magnitude response – Location of Poles – filter design.	7
Total number of hours		45